PROJECT PROPOSAL

Geographic Patterns in Diabetes and Related Risk Factors in the United States

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PURPOSE

The purpose of this proposal is to introduce my project topic and justify my choice of a real-world public health dataset. For this project, I will use the CDC PLACES dataset to examine geographic patterns in diabetes and related chronic disease risk factors across U.S. counties. The proposal provides background information about the dataset, explains why diabetes was selected as the public health topic, and describes how the dataset can be used for public health informatics analysis and visualization.

1. Which real-world dataset did you pick?

For my final project, I selected the **CDC PLACES: Local Data for Better Health, County Data, 2025 release** from the Centers for Disease Control and Prevention (CDC) Data website. PLACES provides model based health estimates for local geographic areas throughout the United States, including counties. The dataset covers all 50 states and the District of Columbia and provides information that can be used to examine differences in health outcomes and risk factors across communities (CDC, n.d.)

The dataset contains 40 measures covering several public health categories, including health outcomes, preventive services, chronic disease related health risk behaviours, disabilities, health status, and health related social needs. Because the data are available at the county level, the dataset can be used to examine geographic patterns and differences in public health indicators across the United States.

For this project, I plan to focus primarily on diabetes and related risk factors, including obesity, physical inactivity, smoking, high blood pressure, and high cholesterol. These variables will

allow me to explore how diabetes prevalence varies geographically and how it relates to other important chronic disease indicators.

2. **Why did you decide to work on this topic?**

I selected diabetes because it is an important public health issue that can be examined through multiple related health indicators. Diabetes is not an isolated health condition; it is associated with several behavioural and chronic disease risk factors. Examining diabetes together with measures such as obesity and physical inactivity can provide a broader understanding of the factors associated with differences in diabetes prevalence across communities.

I am also interested in this topic because it demonstrates how public health informatics can be used to transform large amounts of health data into information that can support public health planning. Rather than looking at individual cases, the PLACES dataset allows health indicators to be examined across geographic areas. This makes it possible to identify communities with higher estimated levels of diabetes and related risk factors.

From a public health informatics perspective, geographic analysis can help public health professionals identify areas that may need additional prevention resources or targeted interventions. Tableau visualizations can make these geographic patterns easier to understand by presenting the data through maps, charts, and comparisons.

The topic is also appropriate for this project because the same dataset can be used throughout the different stages of the assignment. The data can support visualizations, interpretation of public health patterns, and recommendations for prevention and surveillance. This will allow the project to maintain a consistent connection between the dataset, analysis, visualization, and public health implications.

3. **What background information did you find using the dataset homepage?**

According to the CDC dataset homepage, PLACES provides model based estimates for local areas in the United States. The 2025 county release includes county level estimates and covers the 50 states and the District of Columbia. The dataset is provided by the CDC's National

Centers for Chronic Disease Prevention and Health Promotion, Division of Population Health, Epidemiology and Surveillance Branch (CDC, n.d.).

The dataset includes 40 measures across multiple categories. These measures include 12 health outcomes, 7 preventive service measures, 4 chronic disease related health risk behaviours, 7 disability measures, 3 health status measures, and 7 health related social needs measures. The CDC explains that these estimates can be used to identify emerging health problems and help develop and implement targeted public health prevention activities (CDC, n.d.).

The geographic structure of PLACES is particularly useful for this project because the data can be examined at the local level rather than only at the national or state level. County level information allows the project to investigate whether diabetes and related risk factors vary across different communities.

The dataset is also appropriate for visualization because geographic information can be incorporated into Tableau maps and other visualizations. For example, a Tableau map could display estimated diabetes prevalence by county, while bar charts could compare counties with different levels of obesity or physical inactivity. These visualizations can help communicate geographic patterns that may be more difficult to recognize in a spreadsheet.

Overall, the PLACES dataset provides a useful foundation for examining geographic differences in diabetes and related public health indicators. Its broad geographic coverage, multiple health measures, and local level estimates make it feasible to use the same dataset for the visualization, final paper, and presentation portions of this project.

REFERENCES

CDC, 2025. (n.d.). *PLACES: Local Data for Better Health, County Data, 2025 release*. https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-County-Data-20/swc5-untb